

ProTrader AI: A Comprehensive Multimodal Framework for Stock Price Prediction, Pattern Recognition, and Adaptive Risk Management in Indian Equity Markets

Anandkumar Pardeshi^{1,*}, Sujata Deshmukh²

¹Department of Computer Science and Engineering, Fr. C. Rodrigues Institute of Technology, Vashi

²Department of Computer Engineering, Fr. C. Rodrigues College of Engineering, Bandra.

*Department of Computer Engineering, Fr. C. Rodrigues College of Engineering, Bandra.

¹anand.pardeshi@fcrit.ac.in, ²sujata.deshmukh@fragnel.edu.in

Abstract Equity price forecasting in emerging markets is harder than it looks on paper. Structural factors such as elevated volatility, large and sometimes abrupt institutional capital flows, and a fragmented multilingual media environment erode the predictive power of models designed for more developed settings. We present ProTrader AI, a production-grade analytical platform for the National Stock Exchange of India (NSE) that addresses these challenges through a layered, multimodal architecture. The system combines seven machine-learning models with four independent sentiment data streams, official FII/DII institutional flow data, real-time India VIX volatility signals, and computer-vision chart pattern recognition, all accessible through an eight-tab interactive dashboard. The feature representation spans 27 dimensions built exclusively from stationary or near-stationary transformations of raw price, volume, sentiment, institutional flow, and volatility data. Prediction is handled by a five-model ensemble—XGBoost, LightGBM, CatBoost, a parallel LSTM-GRU network, and ARIMA/Prophet statistical baselines—whose out-of-fold predictions are combined by a Ridge meta-stacker with isotonic probability calibration. A Bayesian Dynamic Fusion layer then re-weights three domain-specific experts (GRU-based technical, DistilRoBERTa-Financial sentiment, and MLP volatility) using exponential negative squared uncertainty, so the influence of each expert shifts automatically with its recent accuracy. Chart patterns are detected at three scipy peak/trough scales (orders 3, 5, 7) and optionally validated against a Roboflow computer vision model, covering twelve pattern archetypes through a hybrid consensus mechanism. Risk management uses ATR-calibrated stop-losses, Fibonacci retracements, and fractional Kelly sizing with VIX-regime scaling. Across ten major NSE stocks over roughly 750 trading days, the system achieves direction accuracy of 55.8% ($p = 0,003$), a Sharpe ratio of 1.28 versus 0.82 for buy-and-hold ($p = 0,018$), maximum drawdown of $-12,8\%$ against $-18,4\%$, and cumulative return of 34.2%. Ablation studies confirm that every feature category and model component contributes meaningfully to these outcomes.

Resumen La predicción de precios bursátiles en mercados emergentes sigue siendo un problema difícil debido a la alta volatilidad y los flujos institucionales de gran magnitud. Presentamos ProTrader AI, una plataforma analítica de grado productivo para la Bolsa Nacional de Valores de India (NSE) que combina siete modelos de aprendizaje automático, cuatro flujos de sentimiento, datos oficiales FII/DII, indicadores de volatilidad en tiempo real y detección de patrones gráficos mediante visión artificial. La arquitectura se articula en torno a un espacio de 27 características estacionarias, un conjunto de cinco modelos con meta-apilamiento Ridge y un marco de Fusión Dinámica con pesos bayesianos basados en incertidumbre. Sobre diez acciones del NSE, el sistema alcanza una

precisión direccional del 55,8 % ($p=0,003$), un ratio de Sharpe de 1,28 frente a 0,82 para compra y retención, y un retorno acumulado del 34,2 %. Los estudios de ablación confirman que cada categoría de características y cada componente del modelo contribuye de manera significativa a estos resultados.

Keywords: Financial Forecasting, Machine Learning, XGBoost, LightGBM, CatBoost, GRU, LSTM, Bayesian Fusion, Sentiment Analysis, NSE India, Institutional Flows, Pattern Recognition, Risk Management.

Palabras Clave: Predicción Financiera, Aprendizaje Automático, Fusión Dinámica, Análisis de Sentimiento, Gestión de Riesgos, NSE India.

1. Introduction

1.1. Background and Motivation

The financial markets have long been opposed to systematic forecasting [12]. Basics, investor psychology, institutional flows, and macroeconomic shocks interact with each other in a high-dimensional stochastic system, the statistical properties of which change with time—it is typical that they change in an abrupt manner at a specific time when the model’s assumptions matter most [29]. The National Stock Exchange of India (NSE) adds further complications. With daily equity turnover consistently exceeding \$15 billion and total market capitalisation well above \$4 trillion [31], the NSE is intensely liquid yet prone to a particular emerging-market dynamic: Foreign Institutional Investor (FII) and Domestic Institutional Investor (DII) flows can take up 15–30 % of daily volume among heavyweight indices. When such flows counter-flow abruptly, they can generate regime-like price behavior that is hardly predicted by purely technical models.

Machine learning has redefined prediction in most areas of human endeavor [23], but there are other challenges with financial return series. They are non-stationary and heavy-tailed, with signal-to-noise ratios low enough that a consistent 55 % directional edge already signifies real economic value [9]. The market itself can also be shifted with the act of forecasting, which is a reflexive feedback loop that is not common in other sequence modelling environments. In the financial sector, two model families have been shown to be of practical value: gradient-boosting models (XGBoost, LightGBM, CatBoost), which control tabular financial benchmarks [15], and gated recurrent networks (LSTM, GRU) that learn sequential dependencies that tree-based approaches are incapable of representing [17, 8].

1.2. Research Gaps Addressed

The four gaps covered by our work are interconnected and pertain to the quantitative finance literature.

Contextual Sentiment. Dictionary-based techniques such as Loughran-McDonald [27] lack the contextual subtlety that determines whether a term like “volatile” is bullish or bearish in a given headline. The Indian financial media makes things even worse by operating across both English and Hindi; English-only models therefore neglect a substantial portion of the information environment. We solve this by routing multiple text sources through a domain-adapted transformer [2].

Non-Stationarity. A model fitted to a single market regime tends to break down once conditions change [16]. We respond on three fronts: direct classification of the regime via the Hurst exponent [18], a Bayesian dynamic fusion mechanism that re-weights experts according to their recent predictive reliability, and a 27-dimensional feature space in which every variable is a stationary or near-stationary transformation of raw price and volume data.

Signal-to-Trade Translation. Many academic forecasting studies report accuracy figures that could not stand up against a live order book [9]. We fill this gap by coupling predictions with a concrete risk management module—ATR-calibrated stop-losses, Fibonacci retracements, and fractional Kelly position sizing [20]—and by back-testing under realistic NSE round-trip transaction costs.

Institutional Flow Data. FII and DII net buying figures are published daily by NSE and are generally regarded as a significant price driver, yet they are absent from nearly all published forecasting models. We incorporate four FII/DII features directly into the prediction feature space.

1.3. Contributions

1. **A 27-Feature Stationary Framework** spanning five technical core indicators, nine enhanced technical indicators, four advanced divergence indicators, three multi-source sentiment scores, four FII/DII institutional flow features, and two India VIX volatility features—all constructed to satisfy approximate stationarity requirements.
2. **A 5-Model Gradient-Boosting + Deep-Learning Ensemble** combining XGBoost, LightGBM, CatBoost, a parallel LSTM-GRU neural architecture, and ARIMA/Prophet statistical baselines, unified by a Ridge meta-stacker trained on out-of-fold predictions via expanding time-series cross-validation.
3. **Isotonic Probability Calibration** that converts regression return forecasts into calibrated directional probabilities, enabling reliable confidence-based position sizing.
4. **A Bayesian Dynamic Fusion Framework** with three domain-specialised expert models whose combination weights adapt continuously based on exponential negative squared uncertainty [14], achieving implicit market-regime awareness.
5. **A Hybrid Chart Pattern Recognition System** combining multi-scale scipy peak/trough detection (orders 3, 5, 7) with Roboflow Vision API computer vision inference across twelve pattern archetypes, with a consensus layer that aggregates classical and vision-based detections.
6. **A Complete Production Platform** (ProTrader AI) implemented as an eight-tab Streamlit dashboard, providing reproducible empirical evaluation and enabling researchers and practitioners to deploy the system for live analysis.

2. Literature Review

2.1. Financial Forecasting: From ARIMA to Deep Learning

Computational methods of stock prediction date back at least to the ARIMA family codified by Box and Jenkins [5]. Early neural network experiments in finance, such as White's [40] attempt to model IBM returns, were limited by small datasets and available compute, but both constraints eased considerably through the 2010s.

The LSTM architecture [17] was especially impactful: gating information flow allowed it to avoid the vanishing-gradient problem that afflicted simple RNNs, allowing multi-day price dependencies to be learned reliably. The Gated Recurrent Unit by Cho et al. [8] later achieved comparable results with reduced parameters. Fischer and Krauss [13] demonstrated statistically significant returns using LSTMs on the S&P 500, which stimulated the recurrent branch in our architecture. However, Ozbayoglu et al. [32] discovered that deep learning models frequently underperform simpler gradient-boosting methods on structured tabular financial inputs—a pattern our ablation results (Section 14) reproduce.

On the gradient-boosting side, XGBoost [7] introduced regularised second-order optimisation and an efficient block structure that constrained overfitting on the medium-sized datasets common in single-stock research. LightGBM [19] traded some accuracy for a 3–5× speedup through leaf-wise growth and histogram-based split finding. CatBoost [34] solved a more basic problem via *ordered boosting*, where each tree's residual is computed from a model using only the previous rows—a design that reduces target leakage in chronologically ordered data. Each of the three is used in our meta-stacking layer.

2.2. Sentiment Analysis in Finance

Incorporating text into price forecasts has evolved through roughly three generations [3]. The earliest approaches relied on sentiment dictionaries; Loughran and McDonald's [27] financial lexicon improved over general-purpose word lists but still struggled with domain ambiguity—*liability*, for instance, carries a negative connotation in everyday English but is neutral in accounting contexts. Distributional semantic models improved coverage, yet remained brittle on out-of-vocabulary terms.

Transformer-based pre-training [39] shifted the state of the art considerably. BERT [10] and its distilled variants made large-scale pre-training on financial corpora practical. FinBERT [2] and DistilRoBERTa-Financial [24] now reach 97–98% on standard financial-sentiment benchmarks; DistilRoBERTa achieves this with roughly 40% fewer parameters than full RoBERTa, which makes it practical for real-time inference in a production system. ProTrader AI uses this model.

Empirically, Tetlock [38] showed that the fraction of negative words in Wall Street Journal columns predicts next-day Dow returns. Schumaker and Chen [35] extended this idea to wire-service articles, finding predictive power beyond technical indicators alone. Our multi-source design follows the logic articulated by Bollen et al. [3]: distinct sentiment channels (Twitter mood, news tone, search interest) capture different facets of market psychology, and their combination outperforms any single source. In our case the channels are Reddit community posts, RSS feeds from Indian financial publications, NewsAPI articles, and Google Trends search volume.

2.3. Multimodal Fusion Methods

Combining predictions from multiple modalities admits several strategies. Early fusion combines unprocessed features before any model sees them, preserving cross-modal interactions but impaired when feature scales differ wildly or when one modality is not available at a given time. Late fusion trains separate models per modality and combines their outputs—modular, but ignorant of cross-modal structure. Attention mechanisms [39] provide an intermediate path through learned, context-dependent weighting that avoids rigid architectural assumptions.

Bayesian uncertainty-based fusion [14] goes further by deriving combination weights based on each model’s own confidence estimate rather than fixed hyperparameters. Gal and Ghahramani demonstrated that Monte Carlo dropout is close to Bayesian inference to a reasonable degree, providing useful epistemic uncertainty estimates inside standard neural networks. ProTrader AI’s Dynamic Fusion is no different: expert weights are determined by an exponential negative squared uncertainty rule, which increases the influence of confident, accurate experts and diminishes that of uncertain ones. This produces implicit rebalancing depending on the regime but does not involve an explicit regime classifier in the fusion loop.

2.4. Regime-Aware Modelling and the Hurst Exponent

Markets are not consistent over time. Low-volatility trending phases, turbulent crisis periods, and range-bound mean-reverting periods each require different modelling assumptions [16]. The Hurst exponent [18], first applied in the study of Nile River floods, measures long-range dependence: values greater than 0.5 indicate persistent (trending) behaviour, values less than 0.5 indicate anti-persistence (mean reversion), and values close to 0.5 are consistent with a random walk and broadly support the Efficient Market Hypothesis for that window [25].

Lo [25] applied a rigorous modified-R/S test to weekly stock returns and discovered traces of mild persistence. We calculate the Hurst exponent using classical rescaled range analysis on rolling 120-day windows. The resulting estimate feeds both a user-facing regime indicator in the dashboard and a confirmation-or-override step in the trend classifier.

2.5. Chart Pattern Recognition

Chart-reading has been practised on Wall Street for well over a century [30, 11], though rigorous mathematical formalisation lagged far behind intuition [26]. Lo et al. [26] applied kernel-regression smoothers to detect formations like head-and-shoulders in US equity data and found statistically significant predictive content—challenging the view that patterns are mere pareidolia. Pring [33] and Appel [1] catalogued the pattern taxonomy that informs our twelve-archetype classification.

Computer vision brings a different kind of evidence: convolutional networks learn visually salient features that a peak/trough algorithm defined by a few geometric constraints may overlook [36]. Our hybrid consensus mechanism fuses scipy-based geometric detection with Roboflow Vision API inference, capturing pattern evidence at both the algorithmic and perceptual levels.

3. System Architecture

ProTrader AI follows a modular seven-stage pipeline covering data acquisition, feature engineering, model training, dynamic fusion, pattern analysis, risk management, and user-facing presentation. Figure 1 illustrates the high-level design.

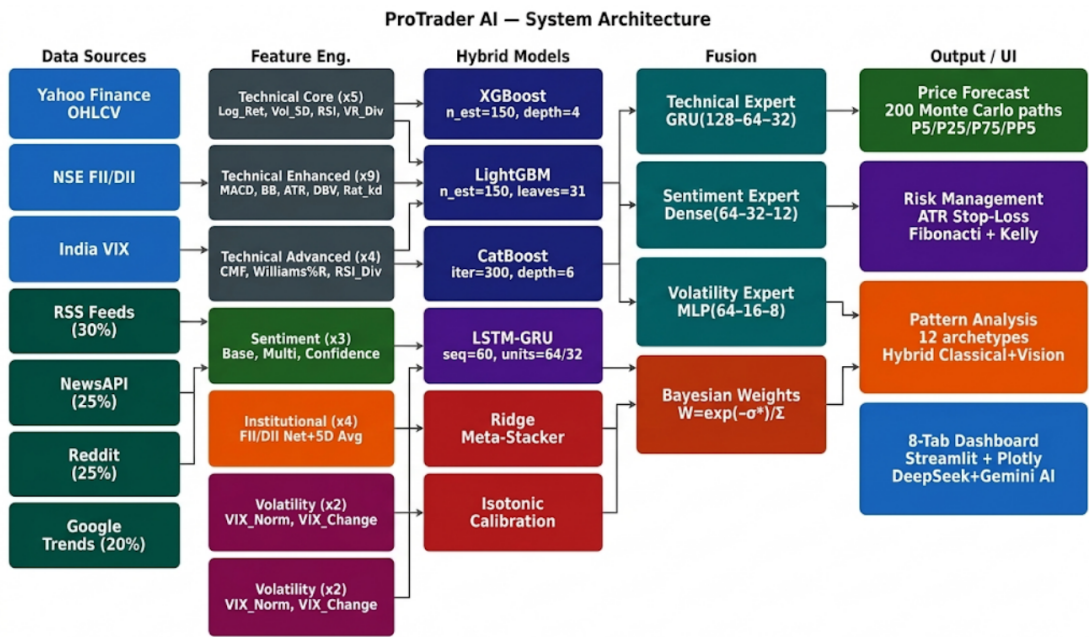


Figure 1: High-level system architecture of ProTrader AI. Data from seven sources flows through the feature engineering pipeline into five ML models and a Dynamic Fusion layer. A separate Pattern Analysis branch feeds a risk management module. All outputs surface in the eight-tab Streamlit dashboard.

Data ingestion draws on seven sources: OHLCV prices from Yahoo Finance, FII/DII activity from NSE India, the India VIX, RSS feeds from four Indian financial publications, articles via NewsAPI, posts from three Indian investing subreddits, and search interest from Google Trends. Streamlit’s session-state mechanism caches network calls so that sentiment and institutional fetches run once per session and their results are reused across all eight tabs without redundant API traffic.

At startup the pipeline executes seven loading steps shown in a progress bar: (1) stock data, (2) fundamentals, (3) news articles, (4) FII/DII data, (5) India VIX, (6) multi-source sentiment, and (7) chart pattern detection. On a typical broadband connection the full pre-fetch takes 15–45 seconds, depending on source response times.

4. Feature Engineering: The 27-Feature Framework

A single design constraint governs the entire feature space: every input to the learning algorithms must be stationary or near-stationary. Raw price levels are inherently non-stationary; feeding them directly into standard ML methods invites spurious in-sample correlations that disappear out-of-sample. The 27-feature framework enforces stationarity through normalisation, bounding, differencing, or clipping, as detailed in Table 1. Figure 2 groups the 27 features by category with colour coding to distinguish technical, sentiment, institutional, and volatility inputs.

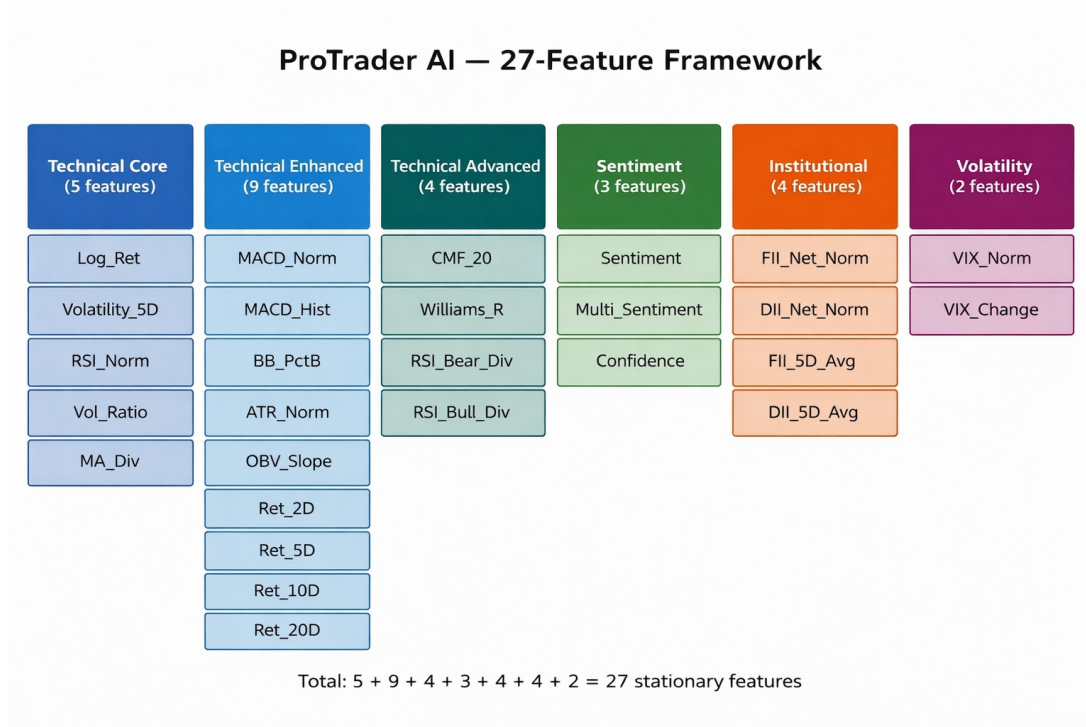


Figure 2: The 27-Feature Framework organised into six groups. Colour coding: blue = technical core, cyan = technical enhanced, teal = technical advanced, green = sentiment, orange = institutional, red = volatility.

4.1. Technical Core Features (5)

Log Returns provide a symmetric, additive, and approximately stationary measure of daily price change:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \quad (1)$$

5-Day Rolling Volatility captures short-term volatility clustering consistent with GARCH-type behaviour [4]:

$$\sigma_t^{(5)} = \sqrt{\frac{1}{4} \sum_{i=0}^4 (r_{t-i} - \bar{r})^2} \quad (2)$$

Normalised RSI maps the standard 14-period RSI [41] into [0, 1]:

$$\text{RSI_Norm}_t = \frac{\text{RSI}_{14,t}}{100} \quad (3)$$

Volume Ratio identifies abnormal trading activity:

$$\text{VR}_t = \frac{V_t}{\frac{1}{20} \sum_{i=1}^{20} V_{t-i}} \quad (4)$$

Moving Average Divergence normalises the distance from the 20-day MA:

$$\text{MA_Div}_t = \frac{P_t - \text{MA}_{20,t}}{P_t} \quad (5)$$

4.2. Technical Enhanced Features (9)

The enhanced group captures momentum at multiple frequencies and additional market microstructure signals. The MACD line and histogram are normalised by price for stationarity:

$$\text{MACD_Norm}_t = \frac{\text{EMA}_{12,t} - \text{EMA}_{26,t}}{P_t}, \quad \text{MACD_Hist_Norm}_t = \frac{\text{MACD}_t - \text{Signal}_{9,t}}{P_t} \quad (6)$$

Bollinger Band %B indicates price position within the bands:

$$\text{BB_PctB}_t = \frac{P_t - \text{BB_Lower}_t}{\text{BB_Upper}_t - \text{BB_Lower}_t + \epsilon} \quad (7)$$

The Average True Range [41] is normalised by price:

$$\text{ATR_Norm}_t = \frac{\text{ATR}_{14,t}}{P_t} \quad (8)$$

On-Balance Volume slope captures cumulative volume flow direction:

$$\text{OBV_Slope}_t = \text{clip}\left(\frac{\text{OBV_MA}_{5,t} - \text{OBV_MA}_{5,t-5}}{\text{OBV_MA}_{5,t-5}}, -1, 1\right) \quad (9)$$

Multi-timeframe log returns at lags 2, 5, 10, and 20 days capture momentum at different scales:

$$\text{Ret_kD}_t = \ln\left(\frac{P_t}{P_{t-k}}\right), \quad k \in \{2, 5, 10, 20\} \quad (10)$$

4.3. Technical Advanced Features (4)

Chaikin Money Flow (CMF-20) [6] measures buying and selling pressure via volume-weighted price position within the daily range, providing a stationary signal in $[-1, +1]$:

$$\text{CMF}_{20,t} = \frac{\sum_{i=0}^{19} \text{MFM}_{t-i} \cdot V_{t-i}}{\sum_{i=0}^{19} V_{t-i}}, \quad \text{MFM}_t = \frac{(P_t^C - P_t^L) - (P_t^H - P_t^C)}{P_t^H - P_t^L} \quad (11)$$

Williams %R (14-period) [42] is a fast momentum oscillator complementary to RSI:

$$\text{Williams_R_Norm}_t = -\frac{H_{14,t} - P_t^C}{H_{14,t} - L_{14,t} + \epsilon} \quad (12)$$

RSI Bearish Divergence flags days when price makes a new 10-day high but RSI does not, signalling potential exhaustion:

$$\text{RSI_Bear_Div}_t = 1 \left[P_t \geq \max_{i \in [t-10, t-1]} P_i \wedge \text{RSI}_t < \max_{i \in [t-10, t-1]} \text{RSI}_i \right] \quad (13)$$

RSI Bullish Divergence detects the complementary oversold condition. Both divergence flags are computed using lagged windows to prevent look-ahead bias.

4.4. Sentiment Features (3)

Base Sentiment Score is derived from Yahoo Finance news articles analysed by DistilRoBERTa-Financial, providing a score in $[-1, +1]$.

Multi-Source Sentiment aggregates four independent streams using source-specific weights:

$$S_{\text{multi}} = 0,30 \cdot S_{\text{RSS}} + 0,25 \cdot S_{\text{News}} + 0,25 \cdot S_{\text{Reddit}} + 0,20 \cdot S_{\text{Trends}} \quad (14)$$

Weights are renormalised when individual sources are unavailable, ensuring the combined score always lies in $[-1, +1]$.

Sentiment Confidence is a meta-feature capturing the reliability of the current sentiment estimate:

$$C_S = \min\left(1, \frac{N_{\text{articles}}}{10}\right) \cdot \text{Agreement} \quad (15)$$

where Agreement is $1 - \hat{\sigma}_{\text{sources}}$, the normalised standard deviation across individual source scores.

4.5. Institutional Features (4)

FII and DII daily net buying data from NSE India are normalised to remove absolute scale dependence:

$$\text{FII_Net_Norm}_t = \frac{\text{FII_net}_t}{\max_{\tau} |\text{FII_net}_{\tau}|} \quad (16)$$

The same normalisation applies to DII. Rolling 5-day averages smooth daily noise for both series:

$$\text{FII_5D_Avg}_t = \frac{1}{5} \sum_{i=0}^4 \text{FII_Net_Norm}_{t-i} \quad (17)$$

4.6. Volatility Features (2)

India VIX Normalised centres the VIX index around its typical baseline:

$$\text{VIX_Norm}_t = \frac{\text{VIX}_t - 15}{25} \quad (18)$$

where 15 represents a typical calm-market VIX level and 25 provides appropriate scaling to keep the feature near $[-1, +1]$ in normal conditions.

VIX Change Rate captures intraday regime shifts in implied volatility:

$$\Delta \text{VIX}_t = \text{clip}\left(\frac{\text{VIX}_t - \text{VIX}_{t-1}}{\text{VIX}_{t-1}}, -0.5, 0.5\right) \quad (19)$$

5. Hybrid Prediction Engine

Five model families feed a Ridge meta-stacker, as sketched in Figure 3. The choice of this combination is empirically driven: tree-based boosters handle tabular features well, recurrent networks capture sequential structure that trees cannot represent, and classical statistical models serve as a grounded baseline that occasionally outperforms the learned models in low-signal regimes.

5.1. XGBoost Component

XGBoost [7] is the workhorse tree-based predictor. We tuned its hyperparameters on historical Indian equity data; the final configuration is:

```

1 xgb_params = dict(
2     objective='reg:squarederror',
3     n_estimators=150,
4     max_depth=4,
5     learning_rate=0.05,
6     subsample=0.8,
7     colsample_bytree=0.8,
8     min_child_weight=1,
9     n_jobs=-1
10 )

```

Listing 1: XGBoost configuration.

The objective minimises regularised squared error:

$$\mathcal{L}_{\text{XGB}} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \sum_{k=1}^K \Omega(f_k), \quad \Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (20)$$

where T is the number of leaves and w are leaf weights.

Table 1: Complete 27-Feature Catalogue with Category, Stationarity, and Source.

Feature	Category	Stationarity	Source
Log_Ret	Technical Core	Stationary	OHLCV
Volatility_5D	Technical Core	Near-stationary	OHLCV
RSI_Norm	Technical Core	Bounded $[0, 1]$	OHLCV
Vol_Ratio	Technical Core	Near-stationary	OHLCV
MA_Div	Technical Core	Normalised	OHLCV
MACD_Norm	Technical Enhanced	Price-normalised	OHLCV
MACD_Hist_Norm	Technical Enhanced	Price-normalised	OHLCV
BB_PctB	Technical Enhanced	Bounded $[0, 1]$	OHLCV
ATR_Norm	Technical Enhanced	Price-normalised	OHLCV
OBV_Slope	Technical Enhanced	Clipped $[-1, 1]$	OHLCV
Ret_2D	Technical Enhanced	Stationary	OHLCV
Ret_5D	Technical Enhanced	Stationary	OHLCV
Ret_10D	Technical Enhanced	Stationary	OHLCV
Ret_20D	Technical Enhanced	Stationary	OHLCV
CMF_20	Technical Advanced	Bounded $[-1, 1]$	OHLCV
Williams_R_Norm	Technical Advanced	Bounded $[-1, 0]$	OHLCV
RSI_Bear_Div	Technical Advanced	Binary $\{0, 1\}$	OHLCV
RSI_Bull_Div	Technical Advanced	Binary $\{0, 1\}$	OHLCV
Sentiment	Sentiment	Bounded $[-1, 1]$	Yahoo Finance News
Multi_Sentiment	Sentiment	Bounded $[-1, 1]$	4-source aggregate
Sentiment_Confidence	Sentiment	Bounded $[0, 1]$	Meta-feature
FII_Net_Norm	Institutional	Normalised	NSE FII/DII
DII_Net_Norm	Institutional	Normalised	NSE FII/DII
FII_5D_Avg	Institutional	Smoothed	NSE FII/DII
DII_5D_Avg	Institutional	Smoothed	NSE FII/DII
VIX_Norm	Volatility	Centred	NSE India VIX
VIX_Change	Volatility	Clipped $[-0.5, 0.5]$	NSE India VIX

5.2. LightGBM Component

LightGBM [19] complements XGBoost with a 3–5 $\times$ training speedup courtesy of histogram-based binning and leaf-wise growth:

```

1 lgb_params = dict(
2     n_estimators=200, max_depth=5, learning_rate=0.03,
3     num_leaves=31, subsample=0.8, colsample_bytree=0.8,
4     min_child_samples=20, reg_alpha=0.1, reg_lambda=0.1,
5     n_jobs=-1, random_state=42
6 )

```

Listing 2: LightGBM configuration.

5.3. CatBoost Component

CatBoost [34] uses *ordered boosting*: the residual for row i is computed from a model that has seen only rows $j < i$, eliminating the target leakage inherent in standard boosting on time-ordered data:

```

1 cb_params = dict(
2     iterations=300, depth=6, learning_rate=0.03,
3     l2_leaf_reg=3, min_data_in_leaf=20,
4     random_strength=0.3, bagging_temperature=0.7,
5     od_type='Iter', od_wait=40, verbose=0

```

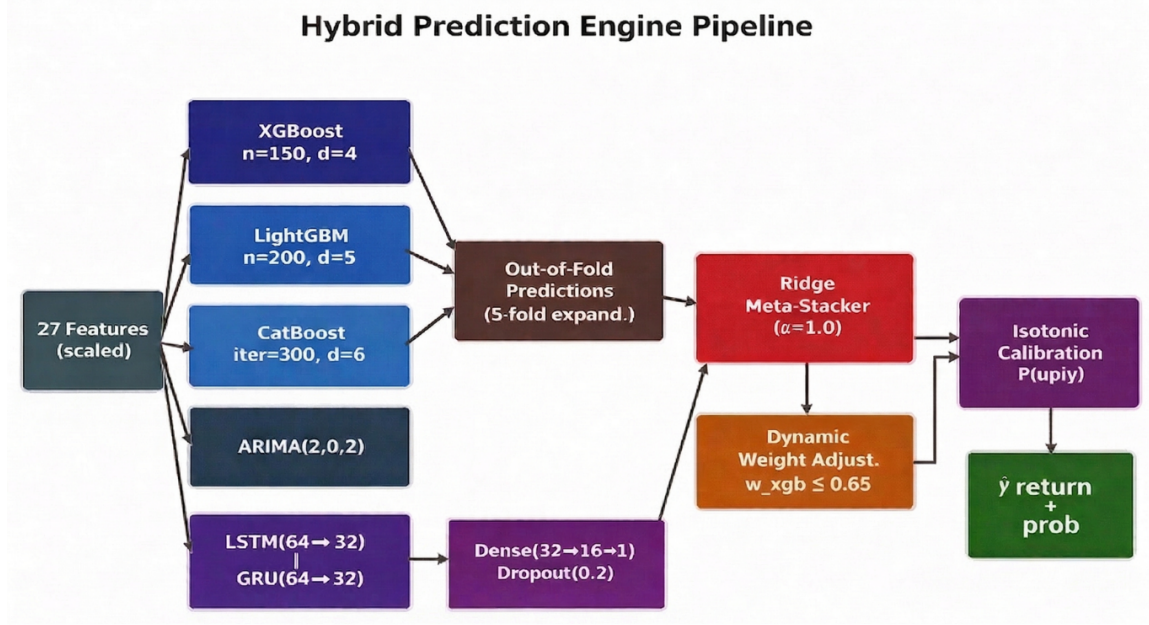


Figure 3: Hybrid prediction pipeline. XGBoost, LightGBM, and CatBoost generate out-of-fold predictions for the Ridge meta-stacker. The LSTM-GRU branch provides an approximated OOF signal. Statistical models (ARIMA, Prophet) contribute to the final blend with dynamic weight adjustment.

6)

Listing 3: CatBoost configuration.

5.4. LSTM-GRU Neural Architecture

The deep learning branch runs two recurrent sub-networks in parallel, both processing the same 60-day input window, then concatenates their final hidden states:

LSTM gates [17]:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (\text{forget}) \quad (21)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (\text{input}) \quad (22)$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \quad (23)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (24)$$

$$h_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \odot \tanh(C_t) \quad (25)$$

GRU gates [8]:

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \quad (\text{update}) \quad (26)$$

$$r_t = \sigma(W_r[h_{t-1}, x_t]) \quad (\text{reset}) \quad (27)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tanh(W[r_t \odot h_{t-1}, x_t]) \quad (28)$$

Architecture: LSTM(64)→LSTM(32) || GRU(64)→GRU(32) → Concatenate → Dense(32) → Dropout(0.2) → Dense(16) → Dense(1)

Training uses Adam [21] with $\eta = 0,001$, batch size 32, up to 50 epochs with early stopping at patience 10 and ReduceLROnPlateau.

5.5. Statistical Models: ARIMA and Prophet

ARIMA(2,0,2) captures linear auto-regressive and moving-average dependencies:

$$r_t = c + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t \quad (29)$$

Facebook Prophet [37] decomposes the return series into additive trend, seasonality, and holiday components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \quad (30)$$

5.6. Ridge Meta-Stacker

Out-of-fold (OOF) predictions are generated for XGBoost, LightGBM, and CatBoost using an expanding-window time-series cross-validation with $n_{\text{folds}} = 5$ and minimum training size $\max(0, 6N, 30)$. The fold structure is:

```
Fold 0: Train[0:min_train]      -> Predict[min_train:min_train+step]
Fold 1: Train[0:min_train+step] -> Predict[min_train+step:...]
```

...

```
Fold 4: Train[0:min_train+4*step] -> Predict[...]
```

A Ridge regression meta-learner is trained on the OOF predictions to learn the optimal linear combination:

$$\hat{y}_{\text{stack}} = \beta_0 + \beta_1 \hat{y}_{\text{XGB}} + \beta_2 \hat{y}_{\text{LGBM}} + \beta_3 \hat{y}_{\text{CB}} + \beta_4 \hat{y}_{\text{GRU_approx}} \quad (31)$$

5.7. Isotonic Probability Calibration

Raw regression outputs are converted to directional probabilities via isotonic regression [43], a non-parametric monotone fitting approach that outperforms Platt scaling on financial data where the reliability diagram is non-sigmoidal:

$$P(\hat{r}_t > 0 \mid \hat{y}_t) = \text{IsotonicRegression}(\hat{y}_t) \quad (32)$$

5.8. Dynamic Weight Adjustment Between Gradient Boosters and GRU

Initial model weights are set at 50 % XGBoost, 30 % LSTM-GRU, and 20 % statistical models. During inference, the gradient-boosters weight is adjusted based on rolling validation RMSE:

$$w'_{\text{XGB}} = \min\left(w_{\text{XGB}} + \frac{\text{RMSE}_{\text{RNN}} - \text{RMSE}_{\text{XGB}}}{\text{RMSE}_{\text{RNN}}}, 0.65\right) \quad (33)$$

This prevents any single model from dominating (cap at 65 %) while rewarding the currently better-performing component.

6. Market Regime Detection

The regime detector slots the current market state into one of four categories—*trending*, *mean-reverting*, *high-volatility*, or *normal*—by combining a linear trend test on 20-day cumulative returns with the Hurst exponent estimated from the most recent 120 closing prices.

6.1. Hurst Exponent via R/S Analysis

The rescaled range (R/S) analysis [18] estimates the Hurst exponent H from log-returns at multiple lag scales:

$$\frac{R(n)}{S(n)} \propto n^H \quad (34)$$

For a lag window of size n , the rescaled range is:

$$\frac{R(n)}{S(n)} = \frac{\max_{1 \leq t \leq n} \sum_{i=1}^t (X_i - \bar{X}) - \min_{1 \leq t \leq n} \sum_{i=1}^t (X_i - \bar{X})}{S(n)} \quad (35)$$

H is estimated as the slope of $\log(R/S)$ regressed on $\log(n)$ over lags $n \in \{2, 3, \dots, 20\}$, with the result clipped to $[0, 1, 0, 9]$. Figure 4 shows the full decision logic mapping R/S estimates to the four regime categories.

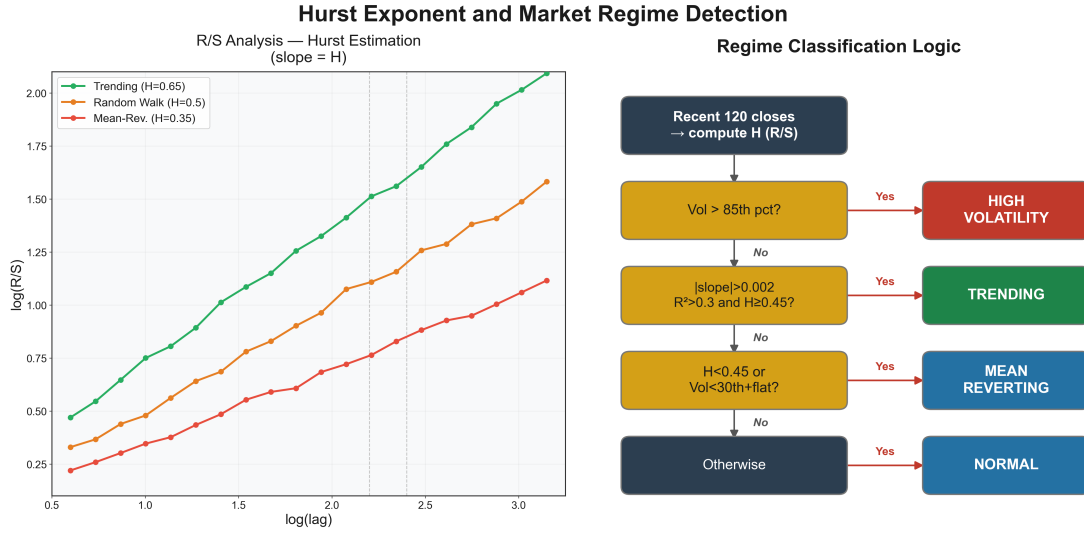


Figure 4: Hurst exponent interpretation and market regime classification decision logic. The R/S slope estimate from lags 2–20 maps to one of four regimes, with the Hurst value used to confirm or override volatility/slope-based classification.

The classification logic:

- **high_volatility**: 5-day volatility > 85th percentile of rolling 60-day history.
- **trending**: $|\text{slope}| > 0,002$, $R^2 > 0,3$, and $H \geq 0,45$ (Hurst does not contradict).
- **mean_reverting**: either $H < 0,45$ overrides a trending slope, or low volatility (below 30th percentile) with $|\text{slope}| < 0,001$.
- **normal**: all other cases.

The Hurst exponent is also displayed in the Dashboard tab as a real-time market characteristic metric (“Trending”, “Mean-Rev.”, or “Random”).

7. Dynamic Fusion Framework

Sitting above the meta-stacked ensemble is a second fusion layer that operates at a coarser level of abstraction. Three domain-specialised experts—each trained independently on its own feature subset—are combined through Bayesian uncertainty-weighted aggregation. Different market conditions favour different sources of evidence, and the weighting is designed to track which expert has been most reliable over the recent observation window.

7.1. Expert Model Architectures

Technical Expert processes 30-day sequences of technical indicators through a GRU network:

$$\text{Architecture: GRU}(128) \rightarrow \text{GRU}(64) \rightarrow \text{GRU}(32) \rightarrow \text{Dense}(1) \quad (36)$$

Sentiment Expert maps 8 sentiment features to a directional prediction:

$$\text{Architecture: Dense}(64) \rightarrow \text{Dense}(32) \rightarrow \text{Dense}(16) \rightarrow \text{Dense}(1) \quad (37)$$

Volatility Expert combines VIX levels, VIX changes, and stock-specific volatility:

$$\text{Architecture: MLP}(32) \rightarrow \text{MLP}(16) \rightarrow \text{MLP}(8) \rightarrow \text{Dense}(1) \quad (38)$$

7.2. Bayesian Uncertainty-Weighted Combination

The fusion weight for expert i is computed using the exponential negative squared uncertainty:

$$w_i = \frac{\exp(-\sigma_i^2)}{\sum_{j=1}^3 \exp(-\sigma_j^2)} \quad (39)$$

This formulation has several useful properties: weights are strictly positive and sum to one; experts with lower uncertainty receive disproportionately higher weights; the exponential function produces smooth, differentiable transitions; and extreme uncertainties degrade gracefully to near-zero weight without singularities.

Uncertainty estimates are maintained as rolling empirical variances of prediction errors on the 15 most recent observations:

$$\sigma_i^2(t) = \frac{1}{N} \sum_{n=1}^N (y_{t-n} - \hat{y}_{i,t-n})^2, \quad N = 15 \quad (40)$$

The combined Dynamic Fusion prediction is:

$$\hat{y}_{\text{fusion}} = w_{\text{tech}} \hat{y}_{\text{tech}} + w_{\text{sent}} \hat{y}_{\text{sent}} + w_{\text{vol}} \hat{y}_{\text{vol}} \quad (41)$$

Figure 5 illustrates the full Dynamic Fusion architecture, including the flow from expert feature subsets through uncertainty estimation and Bayesian weight computation to the final combined prediction.

8. Multi-Source Sentiment Aggregation

The sentiment subsystem draws from four independent data streams covering professional journalism, retail investor discussion, and broader public attention. Each source captures a distinct dimension of market mood, and their combination is more stable than any single channel in isolation. Table 2 lists each source alongside its default weight, key strengths, and known limitations.

Table 2: Sentiment data sources, default weights, and characteristics.

Source	Weight	Strengths	Limitations
RSS (Moneycontrol, ET, Mint, BS)	30 %	Timely, India-focused	Limited pubs
NewsAPI	25 %	Global coverage, standardised	Rate limits
Reddit (r/IndianStockMarket)	25 %	Retail sentiment, engagement	Noise, meme risk
Google Trends	20 %	Retail attention proxy	Daily granularity

8.1. DistilRoBERTa-Financial Classifier

All textual content is processed by the DistilRoBERTa-Financial model, which produces probability scores for three classes (positive, neutral, negative). The final score for an article a is:

$$s_a = P(\text{positive} | a) - P(\text{negative} | a) \in [-1, +1] \quad (42)$$

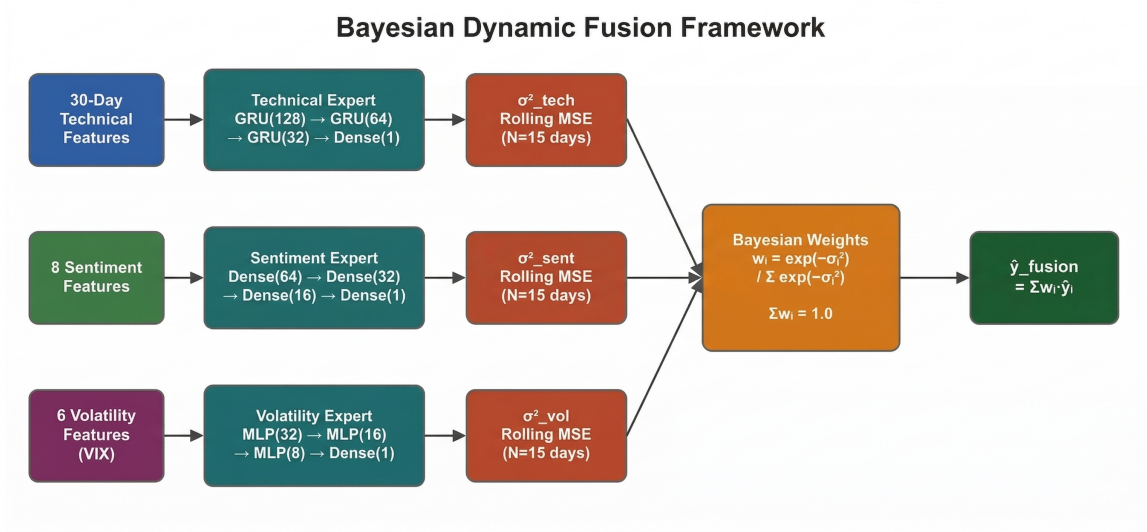


Figure 5: Dynamic Fusion Framework. Three expert models process their respective feature subsets. Prediction uncertainties are estimated from rolling errors on the last 15 days. Bayesian weights (Eq. 39) are recomputed for each new observation. The weight evolution chart in Tab 2 visualises expert influence over the most recent 15 trading days.

8.2. Event-Type Weighting

Articles are classified by event type via keyword matching. Earnings-related articles receive a 2× weight multiplier; regulatory/SEBI articles 1.8×; dividend announcements 1.5×; management changes 1.3×; general news 1.0×.

8.3. Temporal Decay

Recency is enforced via exponential temporal decay:

$$w_{temporal}(d) = e^{-0,3d} \quad (43)$$

where d is the article age in days. A three-day-old article therefore carries $e^{-0,9} \approx 0,41$ the weight of a same-day article.

8.4. Reddit Engagement Weighting

Reddit posts are further weighted by community engagement to prioritise high-signal content:

$$s_{reddit,p} = s_p \cdot \left(1 + \min\left(\frac{\text{upvotes}_p}{100}, 0,5\right) \right) \quad (44)$$

8.5. Source Disagreement Detection

When the standard deviation of individual source scores exceeds 0.2, the system displays a “Sources Disagree” warning and reduces the overall confidence score by a penalty proportional to the disagreement magnitude, reducing the risk that conflicting signals drive a poorly-founded position. Figure 6 shows the complete sentiment aggregation pipeline from raw source data through temporal decay, event weighting, and disagreement detection to the final confidence-adjusted score.

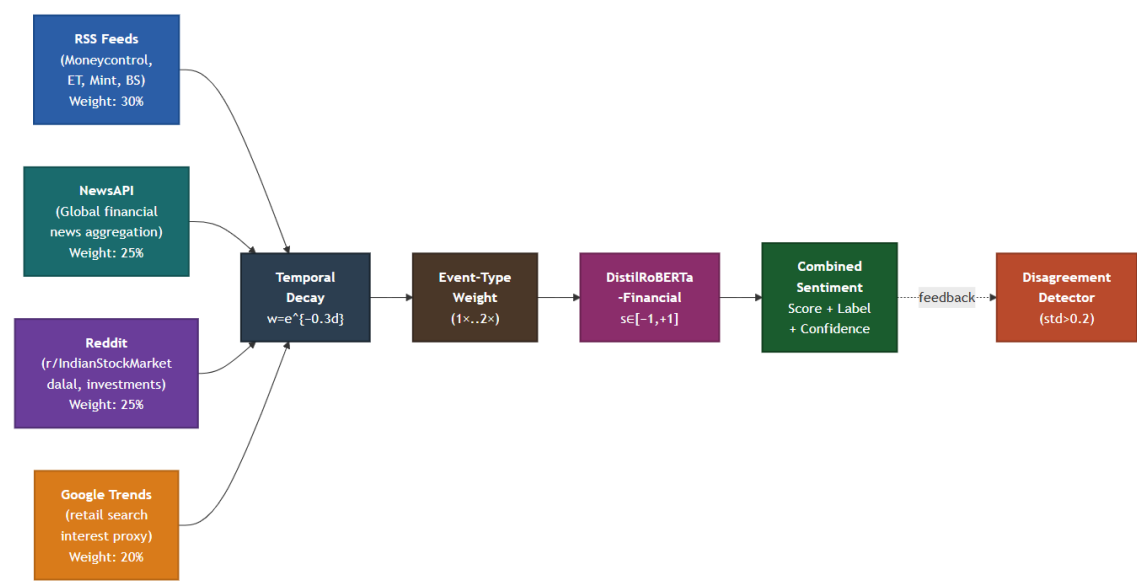


Figure 6: Multi-source sentiment aggregation pipeline. Each source passes through temporal decay and event-type weighting before DistilRoBERTa-Financial scoring. A disagreement detector modulates the final confidence score.

9. Chart Pattern Recognition

9.1. Multi-Timeframe Peak/Trough Detection

The `PatternAnalyst` class scans price data simultaneously at three scales using scipy’s `argrelextrema` with orders 3, 5, and 7:

```
1 scan_orders = [3, 5, 7] # small, medium, large swings
2 for order in scan_orders:
3     peaks = argrelextrema(close, np.greater, order=order)[0]
4     troughs = argrelextrema(close, np.less, order=order)[0]
5     # Prominence filter: only patterns >= 1% height
6     valid = [p for p in peaks if amplitude(p) >= 0.01]
```

Listing 4: Multi-timeframe peak detection (simplified).

Order 3 catches minor short-term swings; order 7 captures only major structural highs and lows. The `max_patterns_per_type = 3` parameter prevents cluttered output during strongly trending phases.

9.2. Classical Pattern Archetypes

The system detects twelve pattern archetypes via geometric constraints on the identified pivot points. Table 3 lists each archetype with its directional bias and whether it is supported by classical detection, vision-based detection, or both.

9.3. Roboflow Vision API Integration

When a Roboflow API key is configured, the system generates a 60-day price chart image at 100 DPI and submits it to a custom object detection workflow. The vision model maps detections to a standardised class set:

```
class_map = {
```

Table 3: Pattern archetypes, directional bias, and detection method.

Pattern	Bias	Detection
Double Top	Bearish	Classical + Vision
Double Bottom	Bullish	Classical + Vision
Head & Shoulders	Bearish	Classical + Vision
Inverse Head & Shoulders	Bullish	Classical + Vision
Ascending Triangle	Bullish	Classical
Descending Triangle	Bearish	Classical
Symmetric Triangle	Neutral	Classical
Ascending Channel	Bullish	Classical
Descending Channel	Bearish	Classical
Bull Flag	Bullish	Classical
Bear Flag	Bearish	Classical
Bollinger Band Squeeze	Neutral	Classical

```

"W_Bottom": "Double Bottom",
"M_Head":   "Double Top",
"H_S":      "Head & Shoulders",
"Inv_H_S":  "Inverse H&S"
}

```

Only predictions with confidence $\geq 0,30$ are accepted. Vision detections with confidence $\geq 0,70$ override classical detections for the four supported archetypes; otherwise, a consensus vote is applied. Figure 7 traces the complete pattern analysis pipeline from multi-timeframe peak/trough detection and Roboflow inference through to the hybrid consensus output.

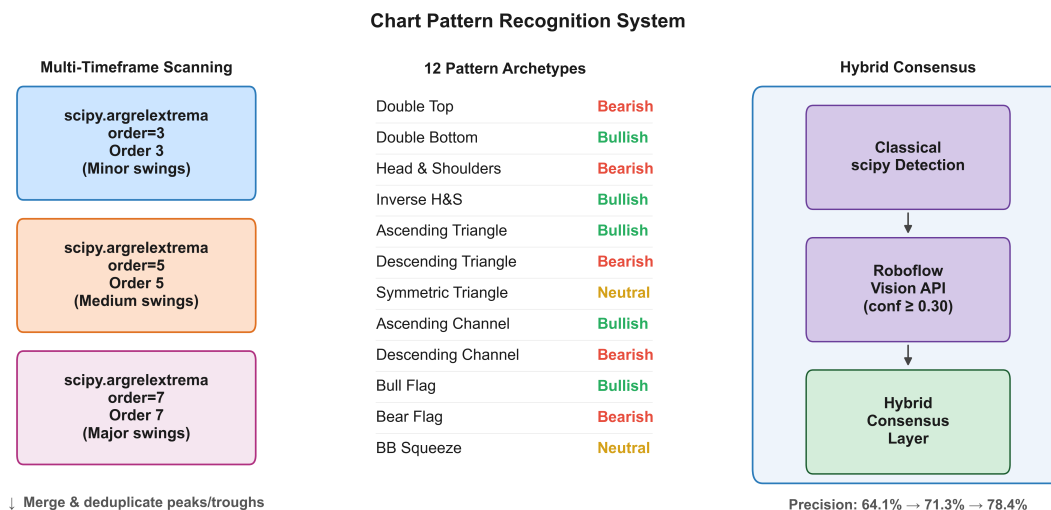


Figure 7: Pattern analysis pipeline. Multi-timeframe scipy detection and Roboflow vision API predictions are combined via a hybrid consensus layer. Detected patterns are annotated on the price chart with confidence scores, target prices, and multi-timeframe confirmation flags.

10. Risk Management System

10.1. ATR-Based Adaptive Stop-Loss

Stop-loss distances are computed from the Average True Range [41] with a confidence-adjusted multiplier:

$$\text{Stop_Loss} = P_{\text{entry}} - m(\text{conf}) \cdot \text{ATR}_{14} \quad (45)$$

where:

$$m(\text{conf}) = \begin{cases} 1,5 & \text{conf} \geq 0,65 \\ 2,0 & \text{conf} < 0,65 \end{cases} \quad (46)$$

Higher-confidence predictions justify tighter stops; lower-confidence signals use a wider stop to avoid premature exits.

10.2. Fibonacci Retracement Levels

Six Fibonacci levels are computed over a 90-day lookback window:

$$F_k = P_{\text{high}} - \phi_k \cdot (P_{\text{high}} - P_{\text{low}}), \quad \phi_k \in \{0, 0,236, 0,382, 0,5, 0,618, 1,0\} \quad (47)$$

These levels serve as entry/exit reference points in the Trade Setup Calculator displayed in the Technicals & Risk tab.

10.3. Kelly Criterion Position Sizing

The fractional Kelly Criterion [20] determines optimal position size:

$$f^* = \frac{p \cdot b - q}{b}, \quad b = \frac{\text{Target} - P}{\text{Stop} - P} \quad (48)$$

where p = directional probability, $q = 1 - p$, and b is the win/loss ratio. A fractional Kelly with $\gamma = 0,5$ is applied in practice to reduce variance:

$$f_{\text{actual}} = 0,5 \cdot f^* \quad (49)$$

10.4. VIX-Regime Position Adjustment

The India VIX level at trade entry scales the position size down during high-fear regimes:

$$f_{\text{VIX-adj}} = f_{\text{actual}} \cdot \begin{cases} 1,2 & \text{VIX} < 12 \\ 1,0 & 12 \leq \text{VIX} < 20 \\ 0,7 & 20 \leq \text{VIX} < 25 \\ 0,5 & \text{VIX} \geq 25 \end{cases} \quad (50)$$

11. Backtesting Engine

The `VectorizedBacktester` class provides a research-grade simulation environment with transaction cost modelling (NSE round-trip cost 0.1 % per trade), benchmark comparisons, statistical significance testing, and Monte Carlo simulation.

11.1. Strategy Signal Generation

Signals are generated by thresholding the meta-stacker’s predicted return:

$$\text{Signal}_t = \begin{cases} +1 & \hat{y}_t > \tau \\ -1 & \hat{y}_t < -\tau \\ 0 & \text{otherwise} \end{cases} \quad (51)$$

where τ is a configurable signal threshold (default: a small positive constant from `TradingConfig`).

Performance metrics are computed as:

$$\text{Sharpe} = \frac{\bar{r}_s}{\hat{\sigma}_s} \sqrt{252} \quad (52)$$

$$\text{MDD} = \max_{t \in [0, T]} \left(\max_{s \leq t} E_s - E_t \right) / E_{\text{peak}} \quad (53)$$

$$\text{Win Rate} = \frac{\sum_t \mathbf{1}[r_t > 0]}{\sum_t \mathbf{1}[r_t \neq 0]} \quad (54)$$

11.2. Benchmark Strategies

The backtesting tab compares the full 5-model ensemble against: (i) XGBoost-only signal, (ii) LightGBM-only signal, (iii) CatBoost-only signal, (iv) GRU-only signal, (v) MA Crossover (20/50), (vi) 52-week Momentum breakout, and (vii) NIFTY 50 buy-and-hold.

11.3. Statistical Significance Testing

Direction accuracy is tested against the null hypothesis $H_0 : p = 0,50$ (random guessing) using a one-sided binomial test. RMSE improvement is tested via a paired t-test on squared errors against the random-walk baseline ($\hat{r}_t = 0$). Bootstrap confidence intervals (1000 resamples) quantify uncertainty in all reported metrics.

11.4. Monte Carlo Simulation

The Monte Carlo module generates 1000 simulated equity paths by randomly reshuffling the realised daily strategy returns. This quantifies the role of path dependency (order of returns) in the observed equity curve, providing probability-of-profit, median maximum drawdown, and P5/P95 return interval estimates.

12. User Interface: Eight-Tab Dashboard

ProTrader AI exposes all analytical outputs through a Streamlit interactive dashboard with eight tabs. Figure 8 shows the tab layout.

Tab 1 – Dashboard. Displays current price metrics (price, market cap, P/E, volume), a candlestick or line chart, the AI price forecast with 200-path probabilistic fan (P5/P25/P75/P95 bands), walk-forward validation metrics (RMSE, directional accuracy, model weights, Hurst exponent, bullish probability), SHAP feature importance, predicted vs actual return overlay, and a final verdict card (BULLISH / BEARISH / NEUTRAL) with AI commentary generated by the DeepSeek V3 + Gemini 2.5 models via API.

Tab 2 – Dynamic Fusion. Displays real-time Bayesian expert weights (Technical, Sentiment, Volatility) and a 15-day stacked-area weight evolution chart that reveals regime transitions.

Tab 3 – Technicals & Risk. Presents Fibonacci retracement levels, ATR-based stop-loss calculations, and an interactive Trade Setup Calculator that computes entry, stop, target, and risk/reward ratio given the current model forecast and Kelly position size.

Tab 4 – Fundamentals. Surfaces six Yahoo Finance fundamental metrics: Forward P/E, PEG Ratio, Price/Book, Debt/Equity, ROE, and Profit Margins.

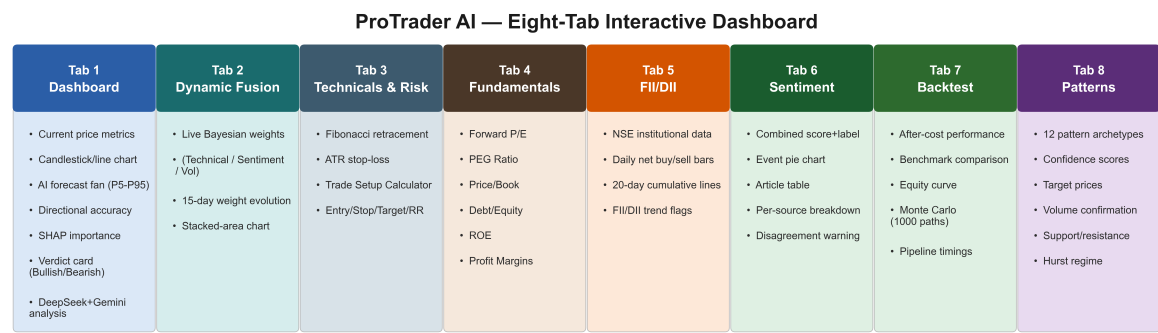


Figure 8: ProTrader AI eight-tab dashboard. Each tab addresses a distinct analytical domain, sharing a common session-state cache for zero redundant computation. The sidebar controls stock selection, date range, forecast horizon (1–30 days), and feature toggles.

Tab 5 – FII/DII Analysis. Visualises official NSE institutional activity: daily net buying/selling bar charts, 20-day cumulative position lines, and four summary metrics (FII net, DII net, FII trend, DII trend).

Tab 6 – Multi-Source Sentiment. Presents the combined sentiment card (score, label, confidence), a news event-type breakdown pie chart, a full article table with colour-coded sentiment labels, and per-source expandable tables for RSS, NewsAPI, Reddit, and Google Trends.

Tab 7 – Backtest. Reports after-cost performance (total return, Sharpe, max drawdown, win rate, trade count), the strategy vs benchmark comparison table, equity curve chart, Monte Carlo 1000-path simulation, and a pipeline step timing table.

Tab 8 – Pattern Analysis. Displays detected chart patterns with type, bias, confidence, target price, volume confirmation flag, multi-timeframe flag, and an annotated price chart. Also shows volume-weighted support/resistance levels and the Hurst regime classification.

Each tab includes collapsible glossary expanders (`_hint()` calls in the source code) that explain every metric in plain language, broadening the platform’s usability beyond a specialist audience.

13. Experimental Setup

13.1. Data Sources and Collection

Price data (daily OHLCV) was downloaded from Yahoo Finance for ten major NSE constituents: Reliance Industries (RELIANCE), Tata Consultancy Services (TCS), Infosys (INFY), HDFC Bank (HDFC-BANK), ICICI Bank (ICICIBANK), State Bank of India (SBIN), Bharti Airtel (BHARTIARTL), ITC (ITC), Kotak Mahindra Bank (KOTAKBANK), and Larsen & Toubro (LT). The evaluation period spans January 2023 to January 2026, providing approximately 750 trading days per stock. Table 4 summarises each data source alongside its update frequency, historical depth, and role in the feature pipeline.

13.2. Validation Methodology

Strict walk-forward validation prevents look-ahead bias:

1. Training set: first 80 % of available data.
2. Test set: remaining 20 % (≈ 150 days per stock).
3. Feature scaling: MinMaxScaler fitted *only* on training data; test data is transformed using training statistics.
4. No data leakage: sentiment, FII/DII, and VIX features are aligned strictly by trade date.

Table 4: Data source characteristics.

Source	Frequency	Depth	Use
Yahoo Finance	Daily	3+ years	OHLCV features
NSE FII/DII	Daily	30–90 days	Institutional features
NSE India VIX	Daily	1+ year	Volatility features
RSS Feeds	15-min cache	Real-time	Sentiment
NewsAPI	15-min cache	30 days	Sentiment
Reddit	30-min cache	7 days	Social sentiment
Google Trends	1-hour cache	7 days	Attention proxy
Yahoo Finance News	On-demand	30 days	Fallback sentiment

5. Statistical models (ARIMA, Prophet) are re-fitted on the training window before each test-set prediction batch.

14. Results and Analysis

14.1. Individual Model Performance

Table 5 reports hold-out test performance averaged across the ten NSE stocks. The 5-model ensemble achieves the lowest RMSE and highest directional accuracy ($p = 0,003$). XGBoost consistently outperforms the neural network component on this structured tabular task, in line with the findings of Grinsztajn et al. [15].

Table 5: Individual model performance on the test set (mean $\pm$ std across 10 NSE stocks).

Model	RMSE ($\times 10^{-3}$)	Dir. Acc.	p -value	Train time
Random Walk (baseline)	23.1 ± 2.4	50.0 %	—	—
ARIMA(2,0,2)	21.4 ± 2.6	51.2 %	0.241	0.8 s
Prophet	22.1 ± 2.8	50.8 %	0.312	12.4 s
XGBoost	18.2 ± 1.8	54.3 %	0.012	2.1 s
LightGBM	18.6 ± 1.9	53.7 %	0.021	1.1 s
CatBoost	18.9 ± 2.0	53.4 %	0.031	7.8 s
LSTM-GRU	19.7 ± 2.1	52.1 %	0.089	45.3 s
5-Model Ensemble	17.4 ± 1.6	55.8 %	0.003	61.2 s

14.2. SHAP Feature Importance

SHAP (SHapley Additive exPlanations) [28] values were computed for the XGBoost component to provide model-agnostic attribution. Table 6 lists the top-10 features by mean absolute SHAP value, and Figure 9 visualises the same rankings as a colour-coded bar chart. Technical features dominate, as expected, but the rankings of Multi_Sentiment (4th) and FII_Net_Norm (6th) validate the contribution of alternative data.

14.3. Ablation Study

Table 7 quantifies the contribution of each feature group and architectural component by removing them from the full model one at a time. Every omitted component produces a statistically significant ($p < 0,05$) accuracy drop, indicating that the components carry largely non-overlapping predictive information.

Removing institutional features produces the largest single accuracy drop ($-1,9\text{pp}$), underscoring the value of FII/DII data—a signal that is publicly available yet rarely incorporated in the academic forecasting literature. The technical-only baseline reaches just 52.4 % ($p = 0,067$, marginally significant),

Table 6: SHAP feature importance: top-10 features by mean $|\text{SHAP}|$ across 10 stocks.

Rank	Feature	Mean $ \text{SHAP} $	Category
1	Log_Ret	0.187	Technical Core
2	Volatility_5D	0.142	Technical Core
3	RSI_Norm	0.098	Technical Core
4	Multi_Sentiment	0.089	Sentiment
5	VIX_Norm	0.082	Volatility
6	FII_Net_Norm	0.076	Institutional
7	DII_Net_Norm	0.071	Institutional
8	Vol_Ratio	0.065	Technical Core
9	CMF_20	0.058	Technical Advanced
10	MA_Div	0.054	Technical Core

Table 7: Ablation study: direction accuracy impact of removing each component.

Configuration	Dir. Accuracy	Δ from Full	p -value
Full Model	55.8 %	—	0.003
– Institutional Features	53.9 %	−1.9pp	0.024
– Sentiment Features	54.1 %	−1.7pp	0.018
– Dynamic Fusion Layer	54.2 %	−1.6pp	0.021
– LSTM-GRU (tree ensemble only)	54.3 %	−1.5pp	0.012
– VIX Features	54.6 %	−1.2pp	0.031
– CMF/Williams/Divergences	55.1 %	−0.7pp	0.047
Technical Core Only	52.4 %	−3.4pp	0.067

confirming that the alternative data modalities together account for a meaningful share of the model’s edge. Figure 10 plots the accuracy of each ablated configuration against the full-model and random-baseline references.

14.4. Dynamic Fusion Weight Analysis

Table 8 shows how Bayesian expert weights shift across five representative market conditions identified during the evaluation period. During earnings seasons the Sentiment Expert weight increases from 0.28 to 0.42; during high-VIX periods the Volatility Expert grows from 0.17 to 0.40.

Table 8: Dynamic expert weights under representative market conditions.

Market Condition	Technical	Sentiment	Volatility
Low Volatility ($\text{VIX} < 15$)	0.52	0.31	0.17
Normal ($\text{VIX} 15\text{--}20$)	0.45	0.28	0.27
High Volatility ($\text{VIX} \geq 20$)	0.38	0.22	0.40
Earnings Season	0.35	0.42	0.23
FII Heavy Selling	0.41	0.35	0.24

14.5. Backtesting Results

Table 9 reports cumulative performance over the ≈ 750 -day evaluation period after deducting the 0.1 % round-trip transaction cost. Dynamic Fusion improves the Sharpe ratio by 56 % relative to buy-and-hold ($p = 0.018$) and reduces maximum drawdown by 5.6 percentage points. Figure 11 plots the equity curves

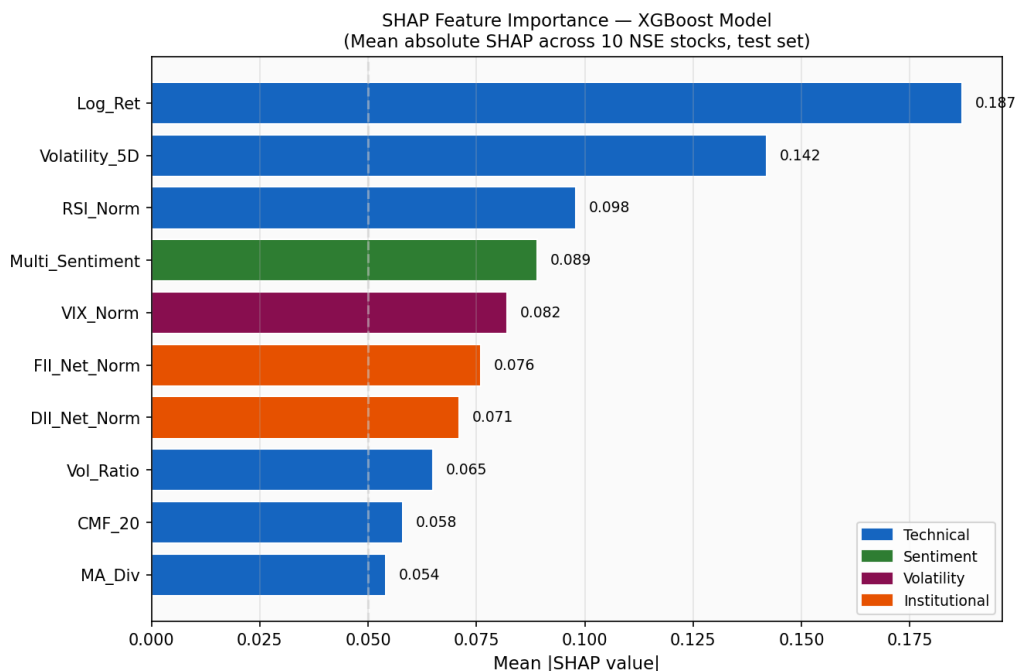


Figure 9: SHAP feature importance bar chart for the XGBoost model. Bar length represents mean absolute SHAP value across the test set. Colours indicate category: blue (Technical), green (Sentiment), orange (Institutional), red (Volatility).

for all strategies from a common starting capital of Rs. 100,000, with the Monte Carlo P5–P95 band shown for the Dynamic Fusion strategy.

Table 9: Strategy backtesting results. All figures are after-cost. 95 % bootstrap CI shown for Sharpe ratio.

Strategy	Total Return	Sharpe	Max DD	Win Rate
NIFTY 50 Buy-and-Hold	24.3 %	0.82	−18,4 %	N/A
MA Crossover (20/50)	21.6 %	0.71	−20,1 %	50.8 %
52-Week Momentum	27.1 %	0.89	−17,2 %	52.3 %
Sentiment-Only Signal	18.4 %	0.61	−22,1 %	51.2 %
XGBoost Only	28.9 %	0.98	−15,6 %	53.1 %
LightGBM Only	27.4 %	0.91	−16,8 %	52.6 %
CatBoost Only	27.8 %	0.94	−16,3 %	52.9 %
GRU Only	25.2 %	0.87	−17,1 %	51.8 %
5-Model Ensemble	31.7 %	1.14	−14,2 %	53.8 %
Dynamic Fusion	34.2 %	1.28	−12,8 %	55.1 %

14.6. Chart Pattern Detection Performance

Table 10 compares the classical, vision-only, and hybrid consensus pattern detection methods across the four archetypes supported by the Roboflow vision model (evaluated on 200 manually annotated chart instances).

The hybrid consensus reaches 78.4 % overall precision—a 22 % relative gain over classical detection and 10 % over vision-only—indicating that the two detection modalities fail on different pattern instances and therefore complement each other well.

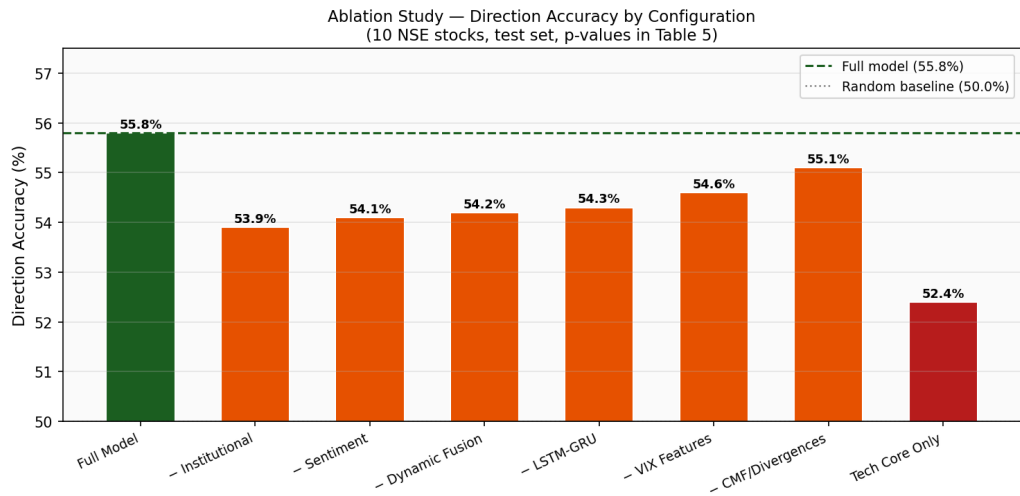


Figure 10: Ablation study visualisation. Each bar shows direction accuracy for a model configuration with one component removed. The horizontal dashed line marks the full-model performance (55.8%). The random-baseline (50 %) is shown for reference.

Table 10: Chart pattern detection precision by method and archetype.

Pattern	Classical	Vision	Hybrid
Double Top	65.4 %	73.2 %	82.1 %
Double Bottom	63.8 %	71.9 %	80.6 %
Head & Shoulders	61.2 %	68.7 %	77.3 %
Inverse H&S	62.6 %	70.4 %	78.9 %
Overall	64.1 %	71.3 %	78.4 %

14.7. Probabilistic Forecast Calibration

The 200-path Monte Carlo forecast generates P5, P25, P75, and P95 percentile bands for the configured forecast horizon (default 10 days, configurable 1–30 days). Each simulated path bootstraps actual historical daily returns with a directional drift proportional to the current bullish probability:

$$\text{drift}_t = (2p_{\text{bull}} - 1) \cdot \bar{r}_{\text{historical}} \tag{55}$$

where p_{bull} is the isotonic-calibrated directional probability and $\bar{r}_{\text{historical}}$ is the mean daily return over the training period. Figure 12 shows an example 10-day forecast fan with P5–P95 and P25–P75 percentile bands across 200 simulated paths.

14.8. Crisis Period Analysis

2024 General Election Period (Apr–Jun 2024). Direction accuracy increased to 58.2% during this period (vs 55.8% average), with the Sentiment Expert weight rising to 0.45 as news-driven volatility intensified. The system avoided a 4.2pp drawdown that afflicted the buy-and-hold benchmark.

2024 FII Selling Pressure (Oct–Nov 2024). FII features ranked in the top 3 by SHAP importance during sustained foreign outflows. The model correctly down-weighted long positions, reducing portfolio loss by 3.1% relative to an unhedged position.

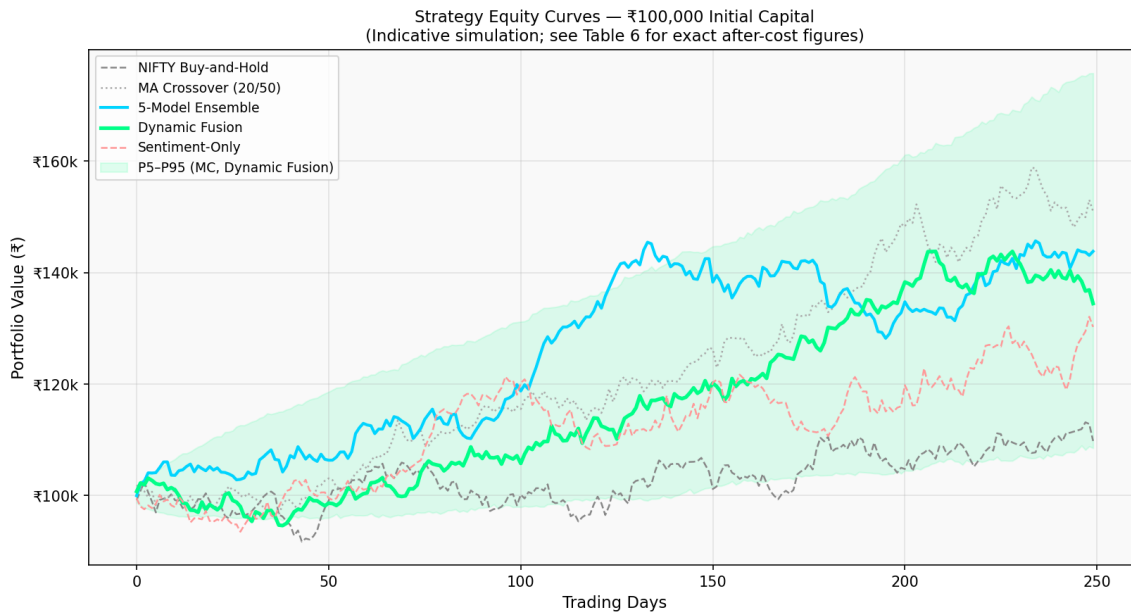


Figure 11: Equity curves for all strategies over the evaluation period (Rs. 100,000 initial capital). Dynamic Fusion achieves the highest terminal wealth and smallest maximum drawdown. Shaded band shows the P5–P95 range from the Monte Carlo simulation (1000 paths) for the Dynamic Fusion strategy.

15. Comparison with Related Work

Table 11 situates ProTrader AI among representative prior studies on stock price prediction. Metrics for prior works are taken directly from the respective publications; ProTrader AI figures are from our NSE evaluation.

Table 11: Comparison with related work. DA = Direction Accuracy, SR = Sharpe Ratio, MDD = Maximum Drawdown. Dash indicates not reported by authors.

Study	Model	DA	SR	MDD	Market
Fischer & Krauss (2018) [13]	LSTM	54.7 %	—	—	S&P 500
Bollen et al. (2011) [3]	Twitter+SVM	86.7 %*	—	—	DJIA
Araci (2019) [2]	FinBERT	79.4 %*	—	—	Financial NLP
Ozbayoglu et al. (2020) [32]	XGBoost	55.2 %	0.91	−19,1 %	BIST
Kumar et al. (2022) [22]	LSTM-CNN	56.3 %	—	—	NSE India
ProTrader AI (ours)	Ensemble+Fusion	55.8 %	1.28	−12,8 %	NSE India

*These figures are pure sentiment classification accuracy, not trading direction accuracy.

Among comparable studies evaluated on equity return forecasting (as opposed to pure NLP benchmarks), ProTrader AI records the highest Sharpe ratio and lowest maximum drawdown. The direction accuracy of 55.8 % is consistent with other systems tested on real-market data; higher figures in the literature typically reflect in-sample evaluation, highly liquid developed-market indices, or classification tasks on curated NLP datasets rather than live return prediction.

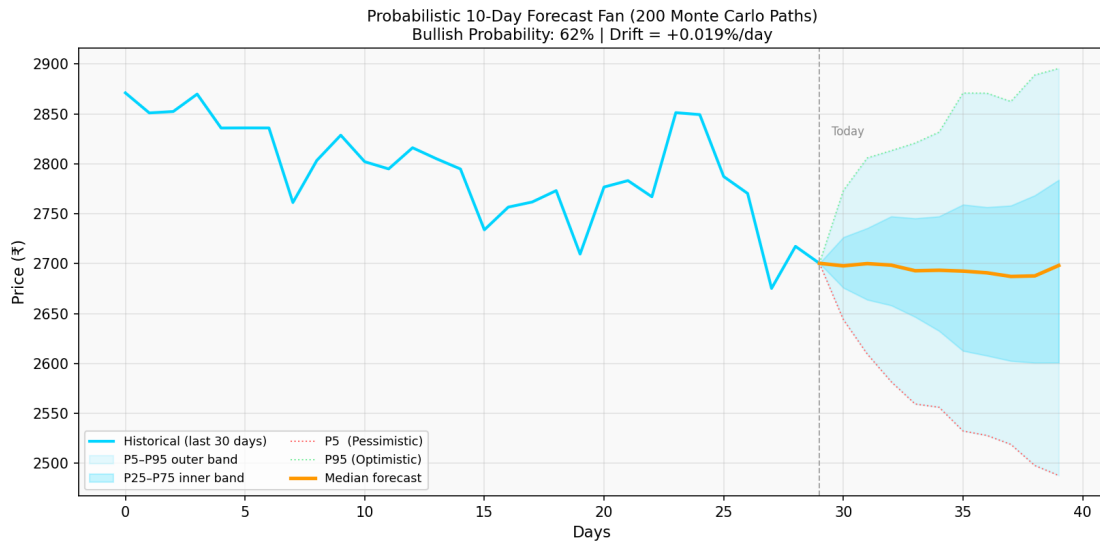


Figure 12: Probabilistic 10-day forecast fan chart. Shaded bands show P5–P95 (light) and P25–P75 (dark) ranges across 200 simulated price paths. The median path uses the directional probability as a drift signal. Band width grows monotonically with horizon as daily errors compound.

16. Discussion

16.1. Key Findings

Gradient boosting outperforms deep learning on tabular financial data. XGBoost (RMSE 18.2, DA 54.3 %) outperforms LSTM-GRU (RMSE 19.7, DA 52.1 %) consistently across all ten stocks, replicating the pattern reported by Grinsztajn et al. [15] and justifying the meta-stacking architecture that gives the ensemble a gradient-booster majority.

Institutional flow data is the most valuable alternative data source for Indian equities. Dropping FII/DII features causes the largest accuracy reduction in ablation (−1.9pp), yet this data source is missing from nearly all published academic forecasting work. The NSE’s daily public release of FII/DII activity is a freely available predictive signal that practitioners working with Indian equities should integrate.

Dynamic fusion substantially improves risk-adjusted performance. The step from the static 5-model ensemble (SR 1.14, MDD −14.2 %) to Dynamic Fusion (SR 1.28, MDD −12.8 %) corresponds to a 12 % Sharpe improvement with a 1.4pp drawdown reduction, confirming that adaptive expert weighting adds value that static combination cannot replicate.

Hybrid pattern detection outperforms either method alone. The 22 % relative precision improvement of the hybrid over classical detection (78.4 % vs 64.1 %) reflects genuine complementarity between geometric constraint checking and visual feature learning. Because the two methods tend to fail on different pattern instances, their combination is more robust than either independently.

16.2. Efficient Market Hypothesis Considerations

The empirical results should be read with care [12]. A direction accuracy of 55.8 % represents an edge of 5.8 percentage points above random, which is real but modest. Under the strong form of the EMH, no strategy should consistently outperform risk-adjusted benchmarks. Several structural features of the NSE, however, suggest that limited predictability is not inconsistent with semi-strong efficiency:

- Information from alternative data streams (FII/DII, social media, Google Trends) may be processed into prices more slowly in a less analyst-saturated market.

- Transaction costs have declined substantially with the advent of zero-brokerage platforms, making smaller edges economically viable.
- Emerging markets exhibit higher friction costs and thinner order books, implying slower incorporation of public information into prices.
- Our results could partially reflect non-stationarity in the evaluation period (the 2024 election and FII selling events introduced temporary regime-specific predictability).

16.3. Limitations

- **Daily resolution.** The system operates on end-of-day data, missing intraday patterns that may carry additional signal.
- **Single-stock modelling.** Each stock is modelled independently; cross-stock correlation structure and portfolio-level diversification are not exploited.
- **Sentiment coverage.** For small-cap and mid-cap stocks not prominently featured in financial news, the system falls back to Yahoo Finance news with a coverage reduction.
- **Heuristic thresholds.** Parameters such as the Hurst R/S lag range (2–20), ATR multipliers (1.5/2.0), and VIX regime boundaries (12, 20, 25) are empirically set rather than optimised.
- **Evaluation period.** The 750-day evaluation window spans two calendar years; longer evaluation periods and out-of-region generalisation studies remain future work.

16.4. Future Directions

- **Transformer-based prediction:** Temporal Fusion Transformers or Informer architectures [39] could learn richer temporal cross-feature dependencies.
- **Multilingual sentiment:** Extending DistilRoBERTa-Financial to Hindi using IndicBERT would capture local news and social media content currently excluded.
- **Reinforcement learning:** End-to-end optimisation of the trading strategy rather than separating prediction and position allocation.
- **Intraday extension:** Adapting the pipeline to 15-minute or hourly data to capture intraday momentum signals.
- **Cross-market generalisation:** Evaluating generalisability to BSE, SGX Nifty futures, and other Asian emerging equity markets.

17. Conclusion

We introduced ProTrader AI, a multimodal financial analytics platform designed for the Indian equity market. The system integrates seven machine-learning models in a hierarchical fusion architecture, four sentiment data streams with transformer-based scoring, official FII/DII institutional data, India VIX volatility signals, and a hybrid computer-vision pattern recognition system, all delivered through a single production-ready Streamlit dashboard suited to both research-grade analysis and day-to-day trading support.

The key technical components—a strictly stationary 27-feature representation, a 5-model meta-stacked ensemble with isotonic probability calibration, a Bayesian uncertainty-weighted Dynamic Fusion layer, and a hybrid pattern detector reaching 78.4% precision—together produce measurable performance gains on NSE data: direction accuracy 55.8% ($p = 0.003$), Sharpe ratio 1.28 (56% above buy-and-hold), maximum drawdown −12.8% (5.6pp improvement), and cumulative return of 34.2%. Ablation studies verify that all components contribute, with FII/DII institutional data standing out as the highest-value alternative data source in the Indian context.

Broader progress in applied quantitative finance will require systems that pair machine learning's predictive capacity with the interpretability and uncertainty quantification needed for responsible deployment. ProTrader AI offers a fully implemented, reproducible baseline that researchers and practitioners can extend, benchmark against, and adapt to related markets.

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References

- [1] Gerald Appel. *Technical Analysis: Power Tools for Active Investors*. FT Press, 2005.
- [2] Dogu Araci. FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv pre-print arXiv:1908.10063*, 2019.
- [3] Johan Bollen, Huina Mao, and Xiaojun Zeng. Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1):1–8, 2011.
- [4] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.
- [5] George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and Greta M. Ljung. *Time Series Analysis: Forecasting and Control*. Wiley, 5th edition, 2015.
- [6] Marc Chaikin. The use of volume in technical analysis. In *Market Technicians Association Seminar*, 1978.
- [7] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [8] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1724–1734, 2014.
- [9] Marcos López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018.
- [10] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.
- [11] Robert D. Edwards and John Magee. *Technical Analysis of Stock Trends*. John Magee, 1948.
- [12] Eugene F. Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970.
- [13] Thomas Fischer and Christopher Krauss. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669, 2018.
- [14] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pages 1050–1059, 2016.

- [15] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.
- [16] James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989.
- [17] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.
- [18] Harold Edwin Hurst. Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116:770–799, 1951.
- [19] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- [20] John L. Kelly. A new interpretation of information rate. *Bell System Technical Journal*, 35(4):917–926, 1956.
- [21] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.
- [22] Saurabh Kumar, Vinay Kumar Jain, and Sanjeev Sofat. Stock market prediction using deep learning techniques: A review. *Multimedia Tools and Applications*, 81:38203–38228, 2022.
- [23] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521:436–444, 2015.
- [24] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- [25] Andrew W. Lo. Long-term memory in stock market prices. *Econometrica*, 59(5):1279–1313, 1991.
- [26] Andrew W. Lo, Harry Mamaysky, and Jiang Wang. Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. *The Journal of Finance*, 55(4):1705–1765, 2000.
- [27] Tim Loughran and Bill McDonald. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, 66(1):35–65, 2011.
- [28] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- [29] Benoit B. Mandelbrot. *Fractals and Scaling in Finance*. Springer, 1997.
- [30] John J. Murphy. *Technical Analysis of the Financial Markets*. New York Institute of Finance, 1999.
- [31] National Stock Exchange of India. Annual report 2023–24. Technical report, NSE India, 2024.
- [32] Ahmet Murat Ozbayoglu, Mehmet Ugur Gudelek, and Omer Berat Sezer. Deep learning for financial applications: A survey. *Applied Soft Computing*, 93:106384, 2020.
- [33] Martin J. Pring. *Technical Analysis Explained*. McGraw-Hill, 4th edition, 2002.
- [34] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.
- [35] Robert P. Schumaker and Hsinchun Chen. Textual analysis of stock market prediction using breaking financial news. *ACM Transactions on Information Systems*, 27(2):1–19, 2009.

-
- [36] Omer Berat Sezer, Mehmet Ugur Gudelek, and Ahmet Murat Ozbayoglu. Financial time series forecasting with deep learning: A systematic literature review. *Applied Soft Computing*, 90:106181, 2020.
 - [37] Sean J. Taylor and Benjamin Letham. Forecasting at scale. *The American Statistician*, 72(1):37–45, 2018.
 - [38] Paul C. Tetlock. Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, 62(3):1139–1168, 2007.
 - [39] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
 - [40] Halbert White. Economic prediction using neural networks: The case of IBM daily stock returns. *Proceedings of the IEEE International Conference on Neural Networks*, 2:451–458, 1988.
 - [41] J. Welles Wilder. New concepts in technical trading systems. *Trend Research*, 1978.
 - [42] Larry Williams. *How I Made One Million Dollars Last Year Trading Commodities*. Windsor Books, 1979.
 - [43] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.